

DARDI PEKA

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PROFESSIONAL SUMMARY

I am a recent graduate of Queen Mary University of London (First Class Honours) and a postgraduate student at Imperial College London with first-hand experience in CFD-driven design, external aerodynamics, and simulation verification. As Head of Engineering for Queen Mary Formula Student, I led a 40+ member team across the full vehicle build, emphasising cross-department integration while delivering designs that balance cost and feasibility. My individual research includes CFD simulations, such as a 2D URANS overset-grid study of tiltrotor wake aerodynamics with rigorous convergence checks. I am keen to apply my skillset with a continuous-improvement approach in engineering roles.

EDUCATION

Imperial College London - London, UK <i>MSc Advanced Aeronautical Engineering</i>	Sept 2026 – Present
Queen Mary University of London - London, UK <i>BEng Aerospace Engineering - First Class - 76.0% Overall (Y1: 76.8%; Y2: 75.8%; Y3: 75.9%)</i>	Sept 2023 – Jul 2026
Westbourne School - Penarth, Wales, UK <i>International Baccalaureate Diploma - HL: Mathematics AA, Physics, Computer Science</i>	Sept 2021 – Jul 2023

EXPERIENCE

Queen Mary Formula Student - London / Towcester, UK	Sept 2023 – Aug 2026
Head of Engineering - Led the engineering team of 7 department heads, 7 deputies and 40+ members, establishing the car-build timeline and engineering Gantt chart to ensure on-time delivery across departments. - Ran weekly management and interdepartmental integration meetings, managing risk, resource prioritisation, performance, and communication with the finance team. - Created the team's Engineering Design Report (EDR) - the competition submission covering the full vehicle concept and design process, structural design and suspension, powertrain choice, cooling system, electronics, and lap time simulation. - Oversaw vehicle assembly and manufacturing, ensuring build quality and correct integration between chassis, suspension, powertrain, drivetrain, and electronics subsystems. - Contributed directly to design decisions across departments, including suspension hardpoint redesign for safer rear mounting, engine bay repackaging for central engine placement, and wiring loom redesign.	Jul 2025 – Aug 2026
Head of Aerodynamics - Led a 6-person team designing and validating the vehicle's aerodynamic package. Ensured CFD knowledge transfer to team members and provided simulation support to other departments. - Used STAR-CCM+ for simulation-driven design of the IC25 front wing, improving downforce by 2.2× while cutting component cost by 30% and saving 4.5 kg (from 11.5 kg) through modifying parametric CAD in SolidWorks. - Led the team through the FSUK 2025 design and cost events (covering for the Chief Engineer position): 9th in design, 5th in cost and 7th overall in the concept class, among 40+ competitors. - Managed the department bill of materials, communication with the finance department, knowledge base and competition documentation.	Jul 2024 – Jul 2025
Senior Team Member - IC Aerodynamics & Lap Time Simulation - Designed parametric front- and rear-wing CAD in SolidWorks, iterating to target aerodynamic performance; co-presented suspension and chassis for the concept class (ranked top 5) with the Head of Suspension. - Modified the EV lap time simulation model using IPG CarMaker and OptimumLap (submission ranked top 7).	Nov 2023 – Jul 2024

Associate Team Member

Sept 2023 – Nov 2023

- Completed a 4-week training programme in SolidWorks (CAD), Ansys Workbench (FEA) and STAR-CCM+/Ansys Fluent (CFD), applied to an external-flow simulation of a previous aero package to secure a senior role in the IC Aerodynamics department.

InvestIN Young Engineer Experience / 2Excel Engineering - London / Lasham, UK

Aug 2022

- Completed InvestIN's 5-day Young Engineer Experience (bridge-building and airfoil design challenges) alongside industry professionals from Rolls-Royce and BAE Systems.
- Assisted with glider assembly and observed launch operations. Gained insight into aircraft maintenance, safety and testing at 2Excel Engineering's MRO facility.

ENGINEERING PROJECTS

Integrated Tiltrotor Design Project - Queen Mary University of London

Sept 2025 – May 2026

Team project developing a modernised Urban Air Mobility tiltrotor based on the Bell XV-15; individual workstream on nacelle conversion mechanics and wake aerodynamics.

- Built a reduced-order 2D URANS CFD methodology in Simcenter STAR-CCM+ to analyse unsteady wake and vortex-shedding behaviour of the nacelle through a 45° conversion sweep.
- Implemented an overset (chimera) grid strategy for large rigid-body rotation, which was selected over mesh morphing via a weighted decision matrix, using a PISO unsteady solver, SST k- ω turbulence and 2nd-order spatial and temporal discretisation.
- Verified numerical credibility through a grid convergence (GCI) study in a simplified simulation case supported by first-cell-height and wall y^+ estimation.
- Ran a five-mesh sensitivity study (40k–700k cells); identified strong drag growth and lift sensitivity beyond ~30° tilt and low-band Strouhal numbers ($St \approx 0.13$ – 0.18). Compared results with relevant bluff-body wake literature, via Q-criterion, λ_2 and FFT analysis.

Supersonic Wind Tunnel Experiment - Queen Mary University of London

Mar 2026 – Apr 2026

Individual coursework: schlieren analysis of a symmetric double-wedge airfoil at $M \approx 1.4$ in the QMUL supersonic wind tunnel.

- Used MATLAB with PIVlab for schlieren frame extraction and image processing; measured shock, expansion and Mach wave angles at 0° and 7.5° incidence with full uncertainty propagation (95% confidence intervals).
- Applied shock-expansion theory, oblique-shock relations, and Prandtl-Meyer functions to calculate surface pressures and force coefficients; identified the breakdown of attached-shock theory at 7.5° where the lower-surface shock detached.

CERTIFICATES & AWARDS

- [CSWA - SolidWorks CAD Design Associate](#)
- [IBM Enterprise Design Thinking Practitioner](#)
- InvestIN Young Engineer Work Experience Certificate
- UKMT Senior Maths Challenge - Bronze

TECHNICAL SKILLS

- CFD & Simulation:** Simcenter STAR-CCM+, Ansys Fluent, external aerodynamics, RANS/URANS with overset grids, SST k- ω turbulence modelling, and verification/convergence studies
- CAD & FEA:** SolidWorks parametric modelling, Ansys Workbench and SIMULIA Abaqus FEA
- Vehicle Simulation:** IPG CarMaker, OptimumLap, OptimumKinematics
- Post-Processing & Analysis:** MATLAB (PIVlab, data processing), FFT / frequency-domain analysis
- Engineering Management:** cost bill of materials (CBOM), programme & Gantt chart planning, design and cost reporting